

Project Wiki: ST3215 Bus Servos & Driver HAT Integration

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1 Overview

This documentation covers the integration, configuration, and software development for the ST3215 Serial Bus Servos controlled via a WaveShare Bus Servo Driver HAT. The system is designed to operate the Tutan-Khamun active gripping mechanism, featuring dual-servo synchronous control, gamepad integration, and sensorless homing capabilities.

2 Hardware Setup

- **Actuators:** 2x ST3215 Serial Bus Servos (Default Baudrate: 1,000,000 bps).
- **Controller:** WaveShare Bus Servo Driver HAT (provides the necessary half-duplex UART conversion).
- **Power Supply:** External 12V DC power supply. *Note: The GND of the 12V supply must be common with the controller board to prevent signal noise.*
- **Input Device:** Logitech Gamepad (interfaced via Linux evdev).

3 System Configuration & Troubleshooting

3.1 Serial Port Permissions (Error 13)

When attempting to communicate with the driver board via USB, you may encounter `PermissionError: [Errno 13] Permission denied: '/dev/ttyACM0'`. This occurs because standard Linux users do not have access to serial interfaces by default.

To resolve this, add your user to the `dialout` group and reboot the system:

```
1 sudo usermod -a -G dialout $USER
2 sudo reboot
```

3.2 Gamepad Input Permissions

Similarly, the `evdev` library requires permission to read raw hardware inputs from the Logitech gamepad. Add your user to the `input` group:

```
1 sudo usermod -a -G input $USER
```

A logout or reboot is required for this group change to take effect.

3.3 USB Latency Timeout (No Status Packet Error)

If the script successfully opens the port and sets the baudrate but returns `[TxRxResult] There is no status packet!`, it is usually caused by the default Linux USB latency timer (16ms) causing a timeout before the servo's reply is processed. Force the latency to 1ms:

```
1 sudo sh -c 'echo 1 > /sys/bus/usb-serial/devices/ttyACM0/latency_timer'
```

4 SDK Modifications

The official WaveShare/Feetech Python SDK contains a critical bug in the `ReadLoad` function, which incorrectly queries the position register instead of the load register. This must be corrected in `sms_sts.py` to enable sensorless homing.

Corrected Code:

```
1 def ReadLoad(self, scs_id):
2     # Target SMS_STS_PRESENT_LOAD_L (Address 60)
3     scs_present_load, scs_comm_result, scs_error = self.read2ByteTxRx(
4         scs_id, SMS_STS_PRESENT_LOAD_L)
5     # ST3215 Load is an 11-bit value. Bit 10 indicates direction.
6     return self.scs_tohost(scs_present_load, 10), scs_comm_result,
7         scs_error
```

Additional telemetry functions (`ReadVoltage`, `ReadTemper`, `ReadCurrent`) were also appended to the SDK to allow for comprehensive CSV logging during operation.

5 Software Architecture

5.1 Control Modes

The Python control script uses the `evdev` library to map Logitech gamepad axes directly to servo velocity profiles.

- **Continuous Rotation (Wheel Mode):** The servos are set to `WheelMode()` to allow for velocity-based control rather than absolute positioning.
- **Dual-Stick Mode:** Left analog stick controls ID 2 (Left Gripper), Right analog stick controls ID 3 (Right Gripper).
- **Single-Stick Mode:** Toggled via the `Triangle/Y` button. The left analog stick drives both servos in opposite directions synchronously to close/open the gripper jaws.

5.2 Sensorless Homing & Limits

Because the mechanism lacks physical limit switches, a sensorless homing sequence was developed using the `ReadLoad` telemetry.

1. Upon startup (or triggered via the `Select` button), the servos drive slowly outward.
2. The script monitors the load at 100Hz.
3. When a sustained current spike ($\text{Load} > \text{HOMING_LOAD_THRESHOLD}$) is detected over 3 consecutive cycles, the script registers a hard stop.
4. The mechanism immediately reverses, finds the opposite hard stop, and calculates a safe operating envelope.
5. An offset (e.g., 3 degrees) is applied to keep the servos away from mechanical binding during normal operation.

5.3 Telemetry Logging

A background CSV logging function records `Timestamp`, `ID`, `Position`, `Speed`, `Load`, `Voltage`, `Temperature` to allow for post-operation mechanical analysis and system tuning.